

# Building and evaluation of a PBPK model for Atomoxetine in CYP2D6 phenotype and activity-score groups

| Version                                                   | evaluation-OSP12.2                                                                                                                                                                          |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| based on <i>Model Snapshot</i> and <i>Evaluation Plan</i> | <a href="https://github.com/Open-Systems-Pharmacology/Atomoxetine-Model/releases/tag/evaluation">https://github.com/Open-Systems-Pharmacology/Atomoxetine-Model/releases/tag/evaluation</a> |
| OSP Version                                               | 12.2                                                                                                                                                                                        |
| Qualification Framework Version                           | 3.5                                                                                                                                                                                         |

This evaluation report and the corresponding PK-Sim project file are filed at:

<https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/>

# Table of Contents

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- [1 Introduction](#)
- [2 Methods](#)
  - [2.1 Modeling strategy](#)
  - [2.2 Data used](#)
  - [2.3 Model parameters and assumptions](#)
- [3 Results and Discussion](#)
  - [3.1 Atomoxetine final input parameters](#)
  - [3.2 Atomoxetine diagnostic plots](#)
  - [3.3 Concentration-time profiles](#)
    - [3.3.1 Model Building](#)
    - [3.3.2 Model Verification](#)
- [4 Conclusion](#)
- [5 References](#)

# 1 Introduction

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Atomoxetine is a selective norepinephrine reuptake inhibitor used for the treatment of attention-deficit/hyperactivity disorder. It is administered orally and is predominantly cleared by CYP2D6-mediated metabolism, which leads to clinically relevant exposure differences between CYP2D6 phenotype and activity-score groups.

This atomoxetine model is intended to describe plasma concentration-time profiles of atomoxetine across CYP2D6 phenotype and activity-score groups. It supports evaluation of CYP2D6 drug-gene interactions and interacting-drug scenarios where atomoxetine exposure is clinically relevant.

The whole-body PBPK model of atomoxetine was developed by [Rüdesheim 2022](#) together with paroxetine and risperidone models to describe CYP2D6 activity score-dependent metabolism. The clinical data include oral single-dose and repeated-dose administration of atomoxetine as solution, capsule, or tablet, CYP2D6 phenotype or activity-score stratified groups, and studies with CYP2D6 inhibition or genetic activity differences.

The presented model includes the following features:

- atomoxetine as the evaluated parent compound,
- CYP2D6-mediated metabolism with activity-score-dependent  $k_{\text{cat}}$  values,
- CYP2C19-mediated metabolism as a pragmatic non-CYP2D6 CYP pathway,
- passive renal filtration,
- oral solution, capsule, and tablet administration in the evaluated clinical studies.

## 2 Methods

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### 2.1 Modeling strategy

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The general concept of building a PBPK model has previously been described by Kuepfer et al. ([Kuepfer 2016](#)). Relevant information on anthropometric and physiological parameters in adults was gathered from the literature and incorporated into PK-Sim as default values for adult simulations ([Willmann 2007](#)).

The applied activity and variability of plasma proteins and active processes integrated into PK-Sim are described in the publicly available PK-Sim Ontogeny Database or otherwise referenced for the specific process.

The atomoxetine model was developed as a single-parent compound model for oral atomoxetine pharmacokinetics. The model by [Rüdesheim 2022](#) described CYP2D6 activity score-dependent metabolism by representing CYP2D6 metabolic capacity with activity-score-dependent  $k_{\text{cat}}$  values. CYP2D6-independent clearance was represented through a CYP2C19 pathway and passive renal filtration.

Clinical studies used for model building covered oral atomoxetine administration and included solution, capsule, and tablet dosing where available. Model-building studies informed oral absorption, CYP2D6-independent clearance in poor metabolizers, and CYP2D6-dependent clearance in extensive metabolizers or activity-score groups. Model verification used independent oral study arms, including CYP2D6 poor-, normal-, intermediate-, and higher-activity groups.

The model-building data covered the key structural questions of the atomoxetine model. Oral solution and capsule data supported absorption parameterization, poor-metabolizer data supported the CYP2D6-independent clearance pathway, and extensive-metabolizer or activity-score data supported CYP2D6-dependent clearance. Verification data were then used to evaluate whether the final model described independent concentration-time profiles without additional study-specific fitting.

The represented processes are CYP2D6-mediated metabolism, CYP2C19-mediated metabolism, plasma protein binding, and passive renal filtration. Atomoxetine metabolites were not explicitly implemented because only limited plasma concentration-time profiles for 4-hydroxyatomoxetine were available. CYP2C19 was used as a surrogate pathway for smaller CYP contributions reported *in vitro* and should be interpreted as a pragmatic clearance pathway rather than as a claim that CYP2C19 is the only non-CYP2D6 metabolic route.

For the CYP2D6 drug-gene interaction model, CYP2D6-dependent clearance was represented with Michaelis-Menten kinetics. CYP2D6  $k_{\text{cat}}$  values were optimized for extensive-metabolizer or activity-score groups, set to zero for poor metabolizers, and scaled for other activity scores using the activity-score framework from [Rüdesheim 2022](#).

The evaluated applications are oral single-dose and repeated-dose applications using solution, capsule, or tablet dosing. The major proteins and processes represented explicitly are CYP2D6, CYP2C19, plasma protein binding, and passive renal filtration. CYP2D6 is the primary mechanistic driver for phenotype and activity-score comparisons, while CYP2C19 and renal filtration account for non-CYP2D6 elimination.

Model performance was assessed with goodness-of-fit diagnostics and concentration-time profiles for the model-building and verification sets. The evaluation emphasizes the separation between CYP2D6 phenotype or activity-score groups because this is the central intended use of the atomoxetine model.

Details about input data are provided in Section 2.2. Details about the structural model and assumptions are provided in Section 2.3.

## 2.2 Data used

### 2.2.1 In vitro and physicochemical data

The table below summarizes the drug-dependent inputs documented for the atomoxetine model.

| Parameter                        | Unit              | Value    | Source                               | Description                                           |
|----------------------------------|-------------------|----------|--------------------------------------|-------------------------------------------------------|
| Atomoxetine MW                   | g/mol             | 255.35   | <a href="#">Zhong 2013</a>           | Compound size used in concentration conversions.      |
| Atomoxetine $pK_a$ base          | -                 | 9.80     | <a href="#">Swain 2012</a>           | Basic ionization constant.                            |
| Atomoxetine solubility, pH 7.4   | mg/mL             | 10.29    | <a href="#">Swain 2012</a>           | Solubility input.                                     |
| Atomoxetine $f_u$                | %                 | 1.30     | <a href="#">Yu 2016</a>              | Plasma binding input.                                 |
| Atomoxetine CYP2D6 $K_m$         | $\mu\text{mol/L}$ | 2.30     | <a href="#">Ring 2002</a>            | CYP2D6 affinity held constant across activity scores. |
| Atomoxetine CYP2D6 $k_{cat}$ EM  | 1/min             | 37.44    | Optimized; <a href="#">Ring 2002</a> | activity-score-dependent CYP2D6 clearance.            |
| Atomoxetine CYP2D6 $k_{cat}$ PM  | 1/min             | 0        | Assumed                              | CYP2D6 poor-metabolizer activity set to zero.         |
| Atomoxetine CYP2C19 $K_m$        | $\mu\text{mol/L}$ | 83.00    | <a href="#">Ring 2002</a>            | CYP2C19 affinity parameter.                           |
| Atomoxetine CYP2C19 $k_{cat}$    | 1/min             | 165.23   | Optimized; <a href="#">Ring 2002</a> | Non-CYP2D6 metabolic clearance surrogate.             |
| Atomoxetine logP                 | -                 | 3.49     | Optimized                            | Distribution input.                                   |
| GFR fraction                     | -                 | 1.00     | Assumed                              | Passive glomerular filtration fraction.               |
| Partition coefficients           | -                 | Diverse  | Calculated                           | Partition coefficient method.                         |
| Cellular permeabilities          | cm/min            | 0.32     | Calculated                           | Cellular permeability method.                         |
| Specific intestinal permeability | cm/min            | 5.23E-05 | Optimized                            | Oral absorption parameter.                            |

## 2.2.2 Clinical data

The evaluation uses 14 observed-data profiles for atomoxetine in peripheral venous blood plasma. The evaluation plan assigns 6 simulations to model building and 8 simulations to model verification.

Model-building clinical data:

| Publication                 | Arm / Treatment / Information used for model building                                                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Nakano 2016</a> | Plasma PK profiles in adults after oral, single-dose administration of 50 mg atomoxetine as solution or capsule with CYP2D6 extensive-metabolizer status. |
| <a href="#">Byeon 2015</a>  | Plasma PK profile in adults after oral, single-dose administration of 40 mg atomoxetine with CYP2D6 AS = 2 status.                                        |
| <a href="#">Kim 2018</a>    | Plasma PK profile in adults after oral, single-dose administration of 20 mg atomoxetine with CYP2D6 AS = 2 status.                                        |
| <a href="#">Sauer 2003</a>  | Plasma PK profiles in adults after repeated oral administration of 20 mg atomoxetine with CYP2D6 extensive- and poor-metabolizer status.                  |

Model-verification clinical data:

| Publication                | Arm / Treatment / Information used for model verification                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Belle 2002</a> | Plasma PK profiles in adults after oral, once daily administration of 20 mg atomoxetine with CYP2D6 extensive-metabolizer status.         |
| <a href="#">Cui 2007</a>   | Plasma PK profiles in adults after oral administration of 40 mg and 80 mg atomoxetine with CYP2D6 AS = 1 status.                          |
| <a href="#">Byeon 2015</a> | Plasma PK profiles in adults after oral, single-dose administration of 40 mg atomoxetine with CYP2D6 AS = 0.5 and AS = 1.25 status.       |
| <a href="#">Kim 2018</a>   | Plasma PK profiles in adults after oral, single-dose administration of 20 mg atomoxetine with CYP2D6 AS = 0.5 status.                     |
| <a href="#">Todor 2016</a> | Plasma PK profiles in adults after oral, single-dose administration of 25 mg atomoxetine with CYP2D6 normal- and poor-metabolizer status. |

## 2.3 Model parameters and assumptions

### 2.3.1 Absorption

The model includes oral solution, capsule, and tablet applications. Oral absorption was described using PK-Sim permeability and formulation settings. The `Specific intestinal permeability` was optimized to 5.23E-05 cm/min based on oral concentration-time data.

The oral solution study by [Nakano 2016](#) provided information-rich data for absorption, while capsule and tablet study arms were used to evaluate oral absorption under clinically used formulations.

No separate dissolution model was introduced for the immediate-release applications. Differences between solution and solid oral applications were represented through the application and formulation settings used for each simulation. This keeps the absorption model compact while preserving the clinically relevant distinction between solution, capsule, and tablet dosing.

Absorption parameterization was evaluated against both model-building and verification study arms. This is relevant because the same intestinal permeability value is used across phenotype and activity-score groups, so absorption should not compensate for differences that are mechanistically attributed to CYP2D6 activity.

### 2.3.2 Distribution

Atomoxetine is highly bound to plasma proteins. A fraction unbound of 1.30% was used as summarized in Section 2.2.1.

Lipophilicity was optimized to logP 3.49. Partition coefficients were calculated with the Berezhkovskiy method, and cellular permeability was calculated by PK-Sim.

The optimized lipophilicity was used together with plasma protein binding to describe tissue distribution and systemic exposure. Because atomoxetine is highly protein bound, changes in fraction unbound would directly affect the unbound concentration available for hepatic metabolism. The value was therefore kept consistent with the documented physicochemical and binding data rather than fitted separately by CYP2D6 group.

Distribution was not stratified by CYP2D6 phenotype. Differences between poor, normal, and higher-activity groups are represented through metabolic capacity, while the same compound distribution assumptions are applied to all adult simulations.

### 2.3.3 Metabolism and Elimination

Two metabolic pathways and passive renal filtration are represented in the model.

- CYP2D6

CYP2D6 is the dominant pathway for atomoxetine clearance in normal metabolizers. CYP2D6 metabolism is represented by Michaelis-Menten kinetics. CYP2D6 poor-metabolizer  $k_{\text{cat}}$  is set to zero, and non-zero activity-score groups use optimized or activity-score scaled  $k_{\text{cat}}$  values.

The same CYP2D6  $K_m$  is used across CYP2D6 groups. Activity-score effects are represented by changes in  $k_{\text{cat}}$ , which is consistent with the activity-score framework used for the CYP2D6 model evaluation. This separates enzyme affinity from phenotype-dependent or activity-score-dependent metabolic capacity.

- CYP2C19

CYP2C19 was implemented as a surrogate pathway for minor non-CYP2D6 CYP contributions reported *in vitro*. CYP2C19  $K_m$  was taken from [Ring 2002](#), and CYP2C19  $k_{\text{cat}}$  was optimized.

This pathway contributes to residual oxidative metabolism in poor metabolizers and prevents the CYP2D6-independent part of clearance from being represented only by renal elimination. The CYP2C19 pathway should therefore be interpreted as a pragmatic model pathway for minor CYP-mediated clearance, not as a claim that CYP2C19 is the only non-CYP2D6 metabolic route.

- Renal elimination

Passive renal filtration is represented with a `GFR fraction` of 1.

Renal elimination is a minor structural pathway relative to CYP-mediated metabolism. It was retained as passive filtration using the model fraction unbound and default adult renal physiology.

## 2.3.4 Automated Parameter Identification

The following parameters were optimized by fitting the model to the data:

| Model Parameter                                         |
|---------------------------------------------------------|
| <code>Lipophilicity</code>                              |
| <code>Specific intestinal permeability</code>           |
| CYP2C19 $k_{\text{cat}}$                                |
| CYP2D6 $k_{\text{cat}}$ for extensive metabolizers      |
| CYP2D6 activity-score-dependent $k_{\text{cat}}$ values |

The optimized parameters were selected to target absorption, distribution, and metabolic-capacity terms that were not fixed by direct source measurements. Parameters with direct physicochemical or binding evidence were not used as free study-specific calibration terms. This supports interpretation of the CYP2D6 activity-score simulations as clearance differences rather than repeated refitting of distribution or absorption.



## 3 Results and Discussion

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The PBPK model for atomoxetine was developed and evaluated with clinical pharmacokinetic data after oral administration. The evaluation covers single-dose and repeated-dose oral administration, solution, capsule, and tablet dosing, and CYP2D6 extensive-, normal-, intermediate-, and poor-metabolizer settings represented by genotype, phenotype, or activity score.

The model-building data supported the oral absorption description, CYP2D6-mediated clearance, CYP2D6-independent metabolism, passive renal filtration, and activity-score-dependent clearance implementation. Model-building studies included solution and capsule data from [Nakano 2016](#), CYP2D6 poor- and extensive-metabolizer data from [Sauer 2003](#), and activity-score stratified data from [Byeon 2015](#) and [Kim 2018](#). Verification included independent oral studies from [Belle 2002](#), [Cui 2007](#), [Byeon 2015](#), [Kim 2018](#), and [Todor 2016](#).

The model quantifies CYP2D6-mediated metabolism, CYP2C19-mediated non-CYP2D6 metabolism, and passive glomerular filtration. The interpretation of model performance requires separate consideration of CYP2D6 poor-metabolizer profiles and profiles with residual or higher CYP2D6 activity. Poor-metabolizer profiles mainly test the non-CYP2D6 clearance pathway and renal filtration. Normal- and higher-activity groups additionally test whether increased CYP2D6  $k_{cat}$  values describe the observed activity-score-dependent exposure decrease.

The next sections show:

1. the final model input parameters for the building blocks: [Section 3.1](#).
2. the overall goodness of fit: [Section 3.2](#).
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: [Section 3.3](#).

The merged GOF diagnostic over the included concentration observations gives a GMFE of 1.28 for atomoxetine. This value shows good population-level agreement across the represented studies, but it should still be interpreted together with the concentration-time profiles because the dataset combines different CYP2D6 activity groups, formulations, and repeated-dose behavior.

[Rüdesheim 2022](#) reported good atomoxetine base-model performance. Across atomoxetine base-model evaluations, the reported mean GMFE was 1.20 for AUClast and 1.18 for Cmax. For plasma concentration values, 88.9% of atomoxetine predictions were within two-fold of the corresponding observed values. These published metrics are consistent with the merged concentration GMFE in [Section 3.2](#).

The DGI ratio evaluation is important for the intended use of this model. The included simulations cover studies by [Cui 2007](#), [Byeon 2015](#), [Kim 2018](#), [Sauer 2003](#), and [Todor 2016](#). [Rüdesheim 2022](#) reported that 5 of 5 AUClast ratios and 5 of 5 Cmax ratios were within the prediction success limits, with GMFE values of 1.25 for AUClast ratios and 1.28 for Cmax ratios.

The concentration-time profiles should be reviewed by CYP2D6 activity group. Since absorption and distribution parameters are shared across activity groups, systematic differences by group would indicate clearance-model limitations rather than formulation-specific adjustment. The current diagnostics support the use of the model for adult oral atomoxetine simulations across the represented CYP2D6 phenotype and activity-score range.

The model is adequate for adult oral atomoxetine simulations within the represented doses, formulations, and CYP2D6 groups. Remaining interpretation should be cautious for unrepresented formulations, pediatric populations, and scenarios where non-CYP2D6 pathways are expected to differ from the adult populations used for model evaluation.

## 3.1 Atomoxetine final input parameters

The final Atomoxetine input parameter table summarizes the drug-dependent model parameters used for the evaluated simulations.

### Compound: Atomoxetine

#### Parameters

| Name                                             | Value                  | Value Origin                                                | Alternative | Default |
|--------------------------------------------------|------------------------|-------------------------------------------------------------|-------------|---------|
| Solubility at reference pH                       | 10.29 mg/ml            | Publication-Literature value in, Swain 2012 [47]            | Measurement | True    |
| Reference pH                                     | 7.4                    | Publication                                                 | Measurement | True    |
| Lipophilicity                                    | 3.4885087539 Log Units | Parameter Identification-Parameter Identification-Optimized | LogP        | True    |
| Lipophilicity                                    | 3.65 Log Units         |                                                             | Mohan 2015  | False   |
| Lipophilicity                                    | 3.81 Log Units         | Publication-Literature value in, Swain 2012 [47]            | ChemAxon    | False   |
| Fraction unbound (plasma, reference value)       | 1.3 %                  | Publication-In Vivo-Literature value in, Yu 2016 [56]       | Measurement | True    |
| Fraction unbound (plasma, reference value)       | 3.9 %                  | Unknown-Watanabe 2018                                       | Calculated  | False   |
| Specific intestinal permeability (transcellular) | 5.23E-05 cm/min        | Parameter Identification-Parameter Identification-Optimized | Optimized   | True    |
| Is small molecule                                | Yes                    |                                                             |             |         |
| Molecular weight                                 | 255.35 g/mol           | Publication-Literature value in, Zhong et al. 2013 [57]     |             |         |
| Plasma protein binding partner                   | Albumin                |                                                             |             |         |

#### Calculation methods

| Name                    | Value           |
|-------------------------|-----------------|
| Partition coefficients  | Berezhkovskiy   |
| Cellular permeabilities | PK-Sim Standard |

## Processes

### Metabolizing Enzyme: CYP2D6-Ring 2002 (4-hydroxylation)

Molecule: CYP2D6

#### Parameters

| Name                                        | Value                        | Value Origin                                                    |
|---------------------------------------------|------------------------------|-----------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 115 pmol/min/mg mic. protein |                                                                 |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein      | Unknown                                                         |
| Km                                          | 2.3 µmol/l                   | Publication-In Vitro-Literature value in, Ring et al. 2002 [39] |
| kcat                                        | 37.4392066898 1/min          | Parameter Identification-Parameter Identification-Optimized     |

### Metabolizing Enzyme: CYP2D6-AS=0

Molecule: CYP2D6

#### Parameters

| Name                                        | Value                      | Value Origin                                                             |
|---------------------------------------------|----------------------------|--------------------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                                          |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                                  |
| Km                                          | 2.3 µmol/l                 | Publication-In Vitro-Same CYP2D6 Km as base model, Ring et al. 2002 [39] |

### Systemic Process: Glomerular Filtration-Assumed

Species: Human

## Parameters

| Name         | Value | Value Origin  |
|--------------|-------|---------------|
| GFR fraction | 1     | Other-Assumed |

## Metabolizing Enzyme: CYP2D6-AS=2

Molecule: CYP2D6

## Parameters

| Name                                        | Value                      | Value Origin                                                                                                            |
|---------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                                                                                         |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                                                                                 |
| Km                                          | 2.3 µmol/l                 | Publication-In Vitro-Same CYP2D6 Km as base model, Ring et al. 2002 [39]                                                |
| kcat                                        | 102.25 1/min               | Parameter Identification-Parameter Identification-Optimized AS=2 baseline, Rüdeshheim et al. 2022, main article Table 2 |

## Metabolizing Enzyme: CYP2D6-AS=0.5

Molecule: CYP2D6

## Parameters

| Name                                        | Value                      | Value Origin                                                                                 |
|---------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                                                              |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                                                      |
| Km                                          | 2.3 µmol/l                 | Publication-In Vitro-Same CYP2D6 Km as base model, Ring et al. 2002 [39]                     |
| kcat                                        | 16.87 1/min                | Other-Manual Fit-CYP2D6 activity-score scaling, Rüdeshheim et al. 2022, main article Table 2 |

## Metabolizing Enzyme: CYP2D6-AS=1.25

Molecule: CYP2D6

## Parameters

| Name                                        | Value                      | Value Origin                                                                                |
|---------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                                                             |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                                                     |
| Km                                          | 2.3 $\mu\text{mol/l}$      | Publication-In Vitro-Same CYP2D6 Km as base model, Ring et al. 2002 [39]                    |
| kcat                                        | 53.68 1/min                | Other-Manual Fit-CYP2D6 activity-score scaling, Rüdeshiem et al. 2022, main article Table 2 |

## Metabolizing Enzyme: CYP2C19-Ring 2002

Molecule: CYP2C19

## Parameters

| Name                                        | Value                       | Value Origin                                                    |
|---------------------------------------------|-----------------------------|-----------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 97 pmol/min/mg mic. protein |                                                                 |
| Content of CYP proteins in liver microsomes | 19 pmol/mg mic. protein     | Unknown                                                         |
| Km                                          | 83 $\mu\text{mol/l}$        | Publication-In Vitro-Literature value in, Ring et al. 2002 [39] |
| kcat                                        | 165.2291488831 1/min        | Parameter Identification-Parameter Identification-Optimized     |

## Metabolizing Enzyme: CYP2D6-AS=1

Molecule: CYP2D6

## Parameters

| Name                                        | Value                      | Value Origin                                                                                 |
|---------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                                                              |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                                                      |
| Km                                          | 2.3 $\mu\text{mol/l}$      | Publication-In Vitro-Same CYP2D6 Km as base model, Ring et al. 2002 [39]                     |
| kcat                                        | 39.7 1/min                 | Other-Manual Fit-CYP2D6 activity-score scaling, Rüdeshheim et al. 2022, main article Table 2 |

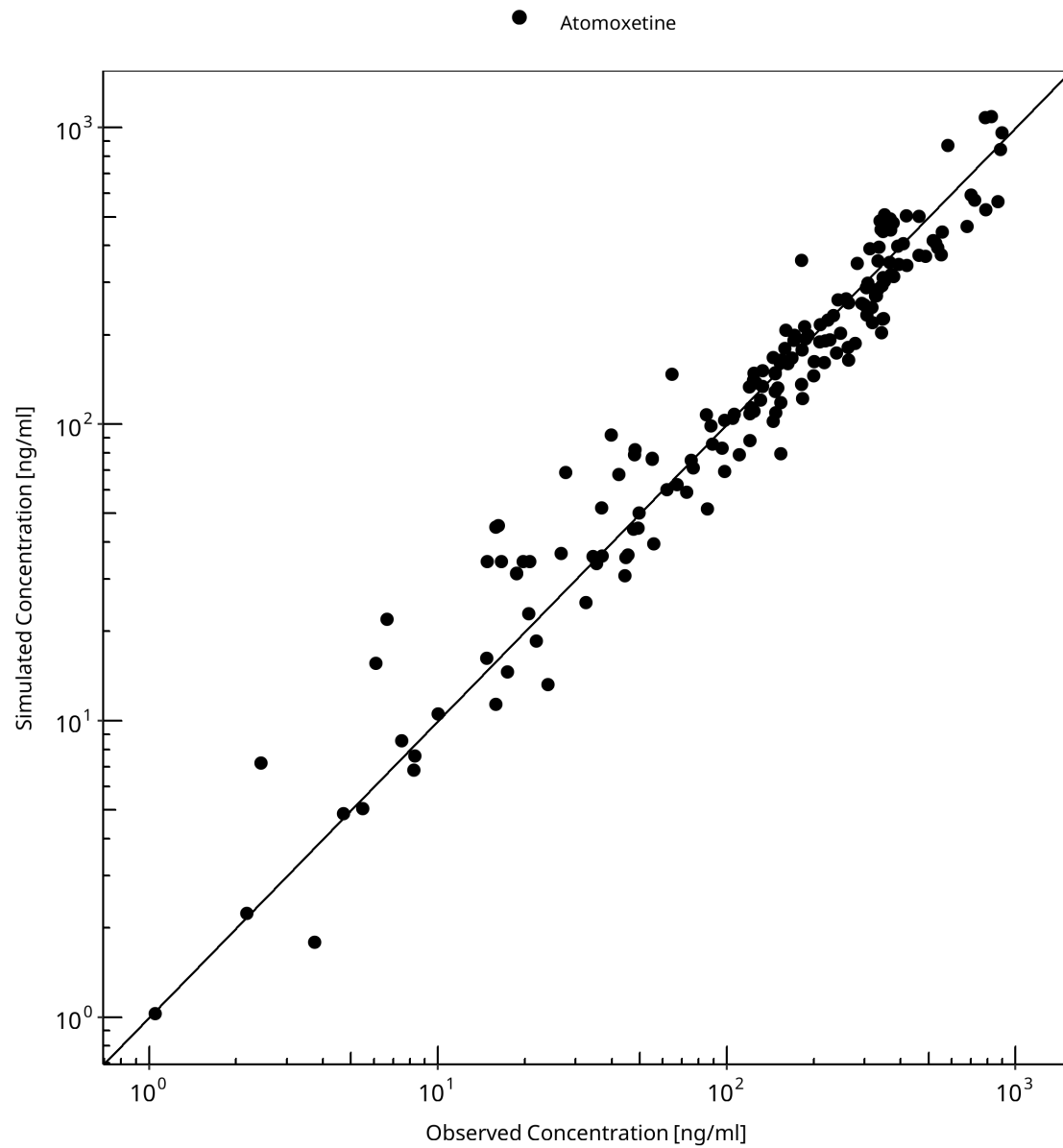
## 3.2 Atomoxetine diagnostic plots

Below you find the goodness-of-fit visual diagnostic plots for PBPK model performance of all data used, as presented in Section 2.2.2 (Clinical data).

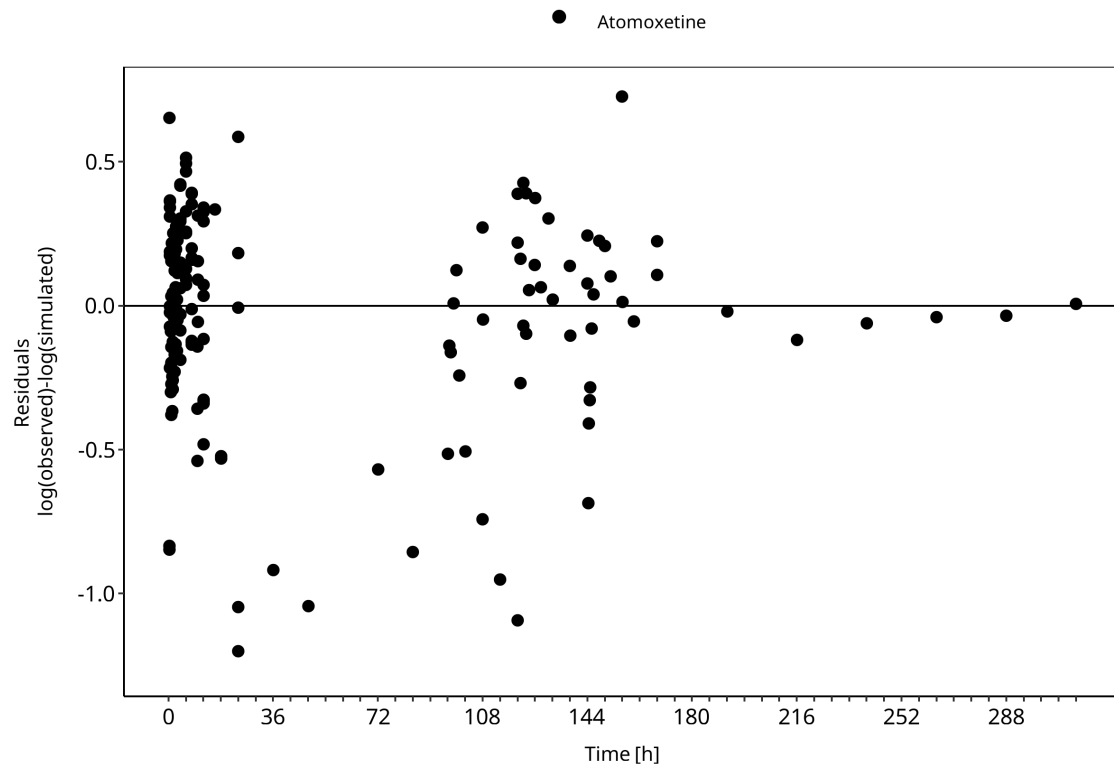
The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

**Table 3-1: GMFE for Atomoxetine observed versus simulated plasma concentration-time data**

| Group       | GMFE |
|-------------|------|
| Atomoxetine | 1.28 |



**Figure 3-1: Atomoxetine observed versus simulated plasma concentration-time data**



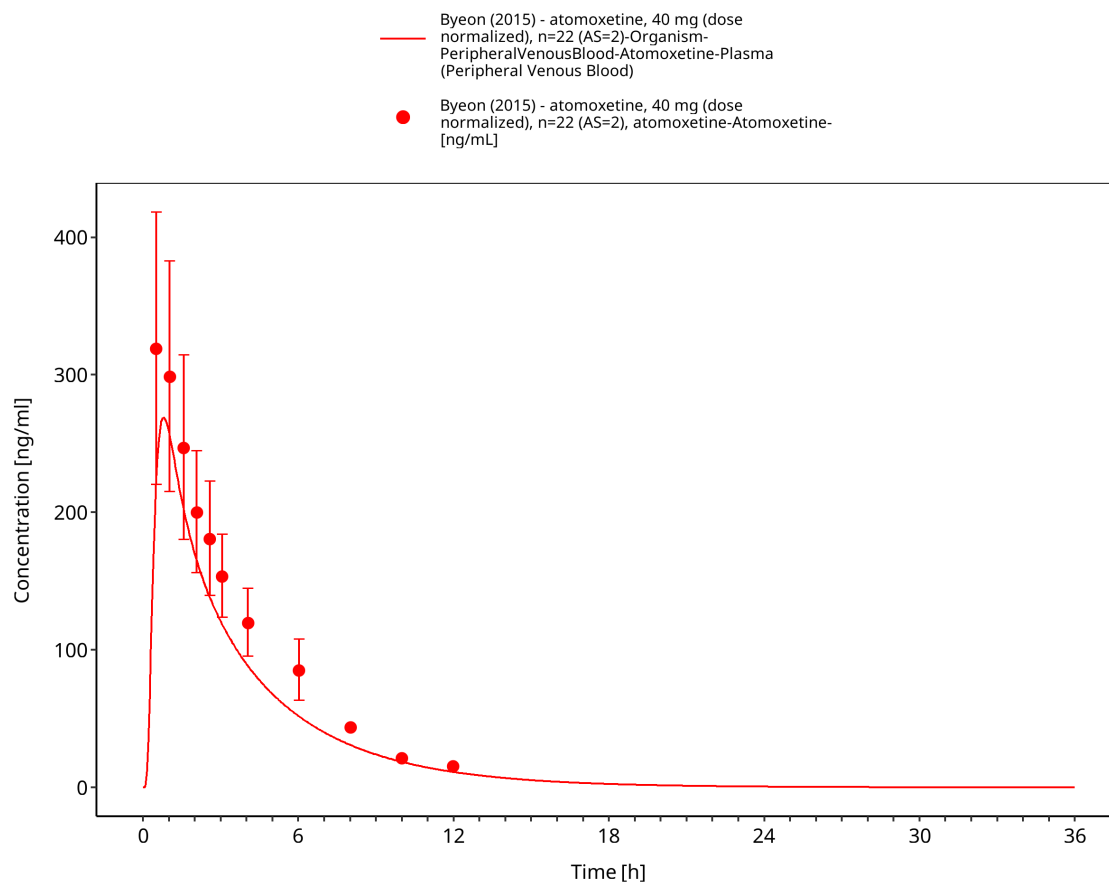
**Figure 3-2: Atomoxetine observed versus simulated plasma concentration-time data**

## 3.3 Concentration-time profiles

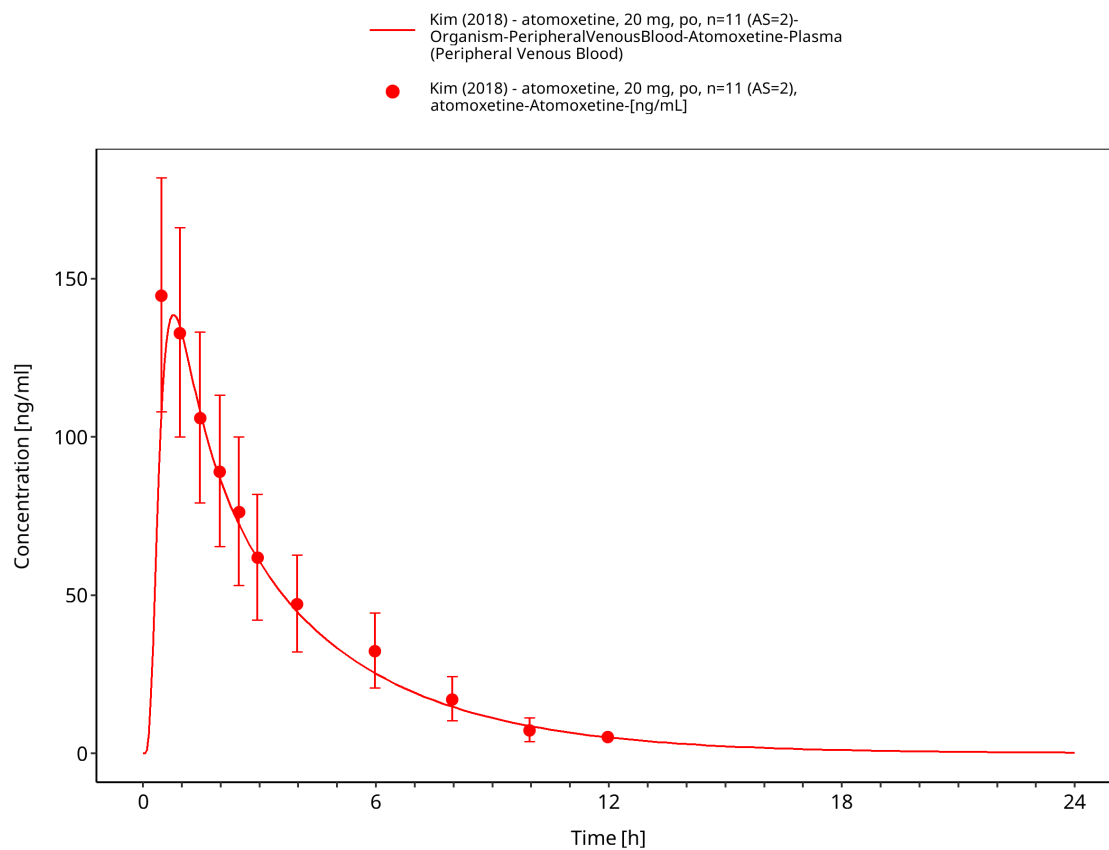
Simulated versus observed concentration-time profiles of all data listed in Section 2.2.2 (Clinical data) are presented below.

### 3.3.1 Model Building

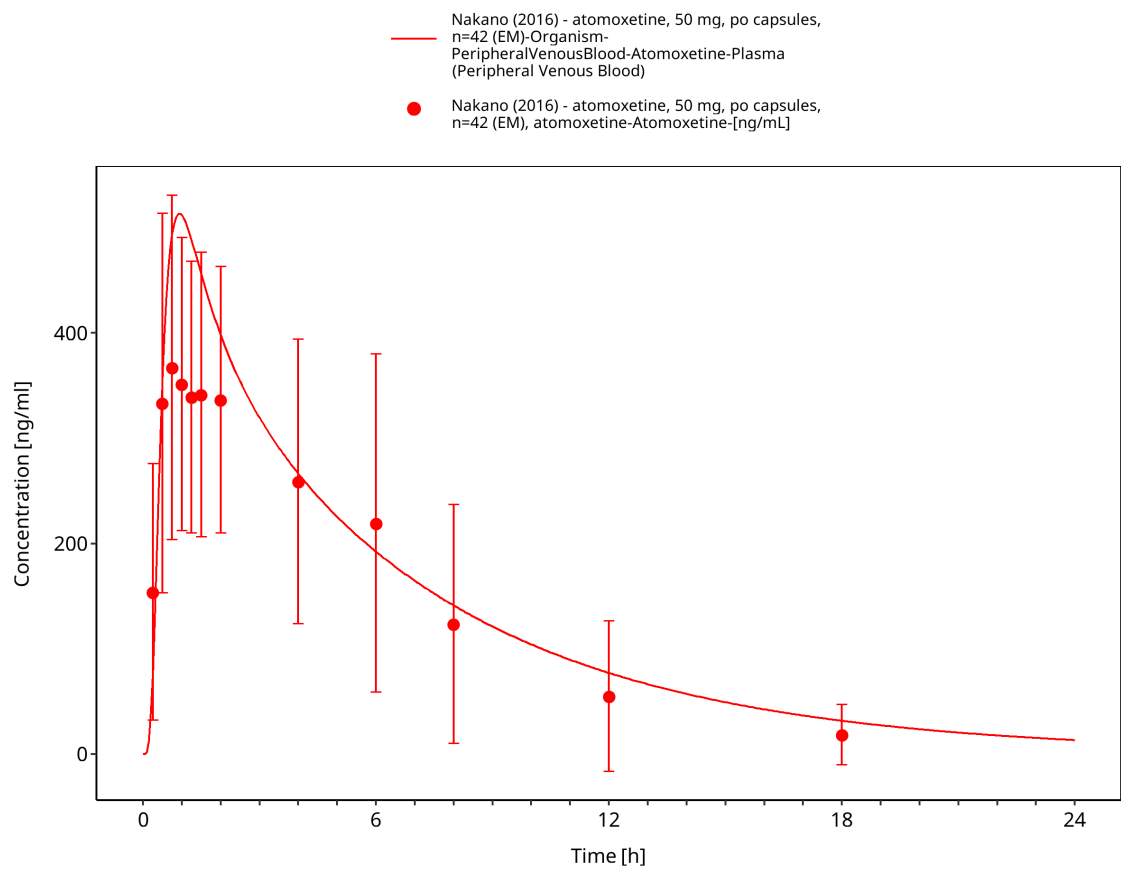




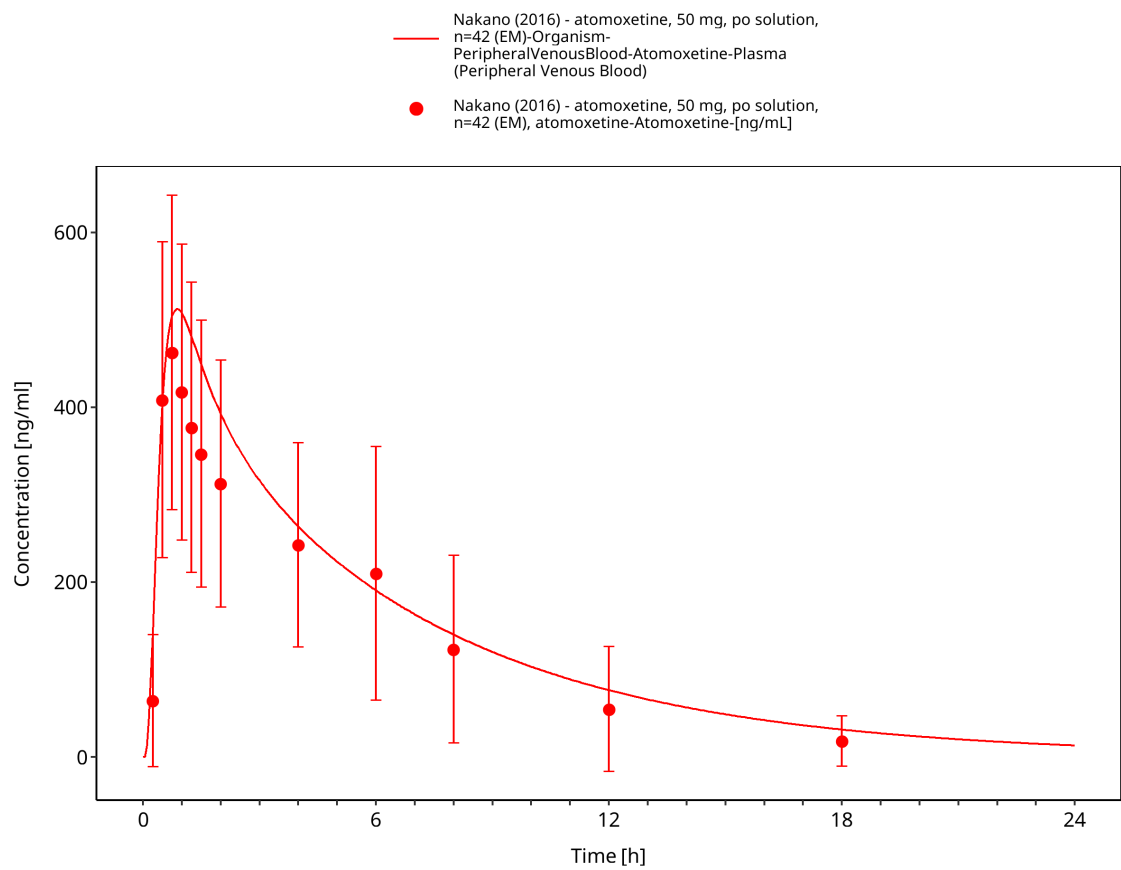
**Figure 3-3: Time Profile Analysis**



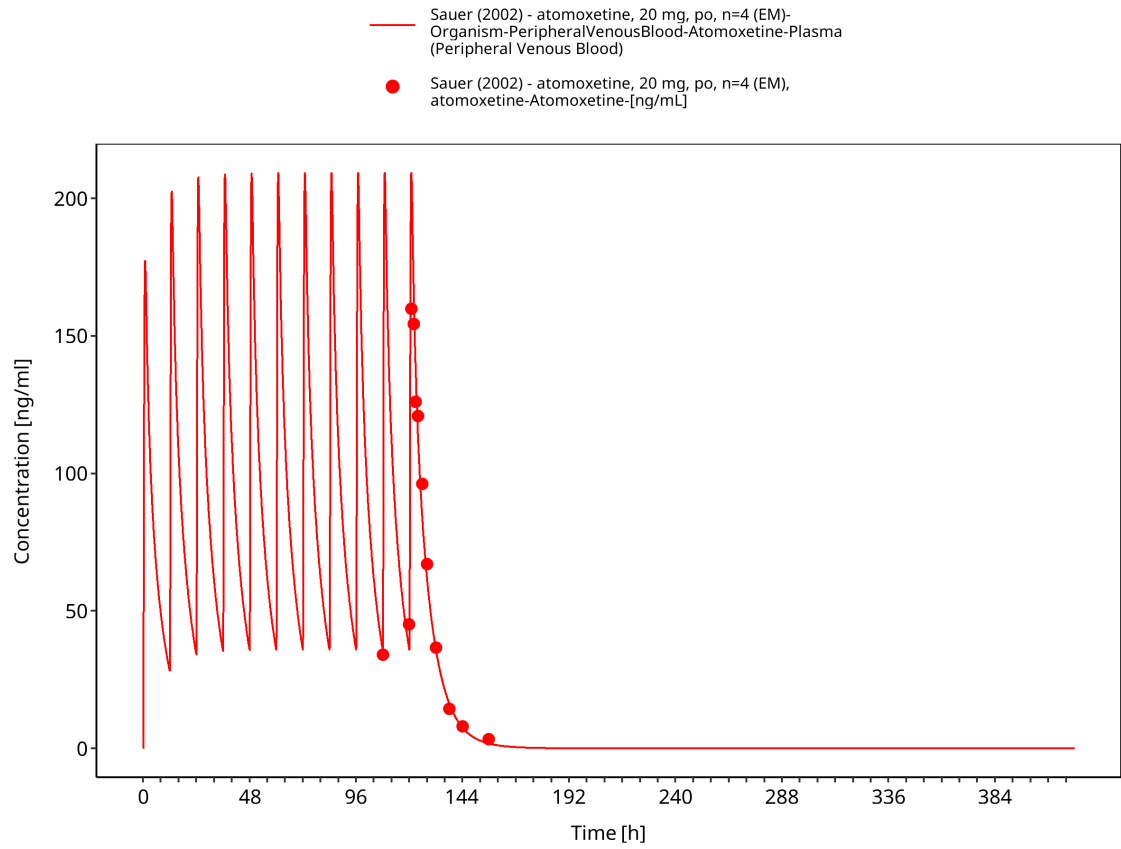
**Figure 3-4: Time Profile Analysis**



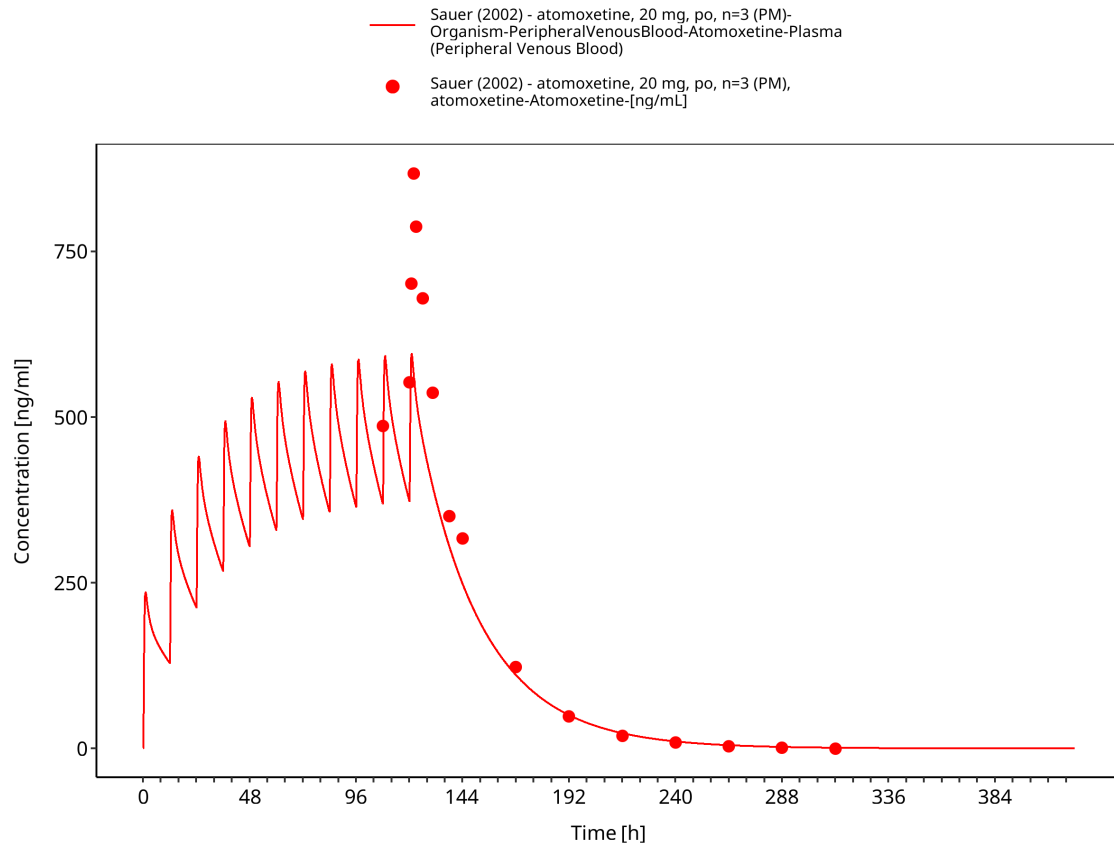
**Figure 3-5: Time Profile Analysis**



**Figure 3-6: Time Profile Analysis**

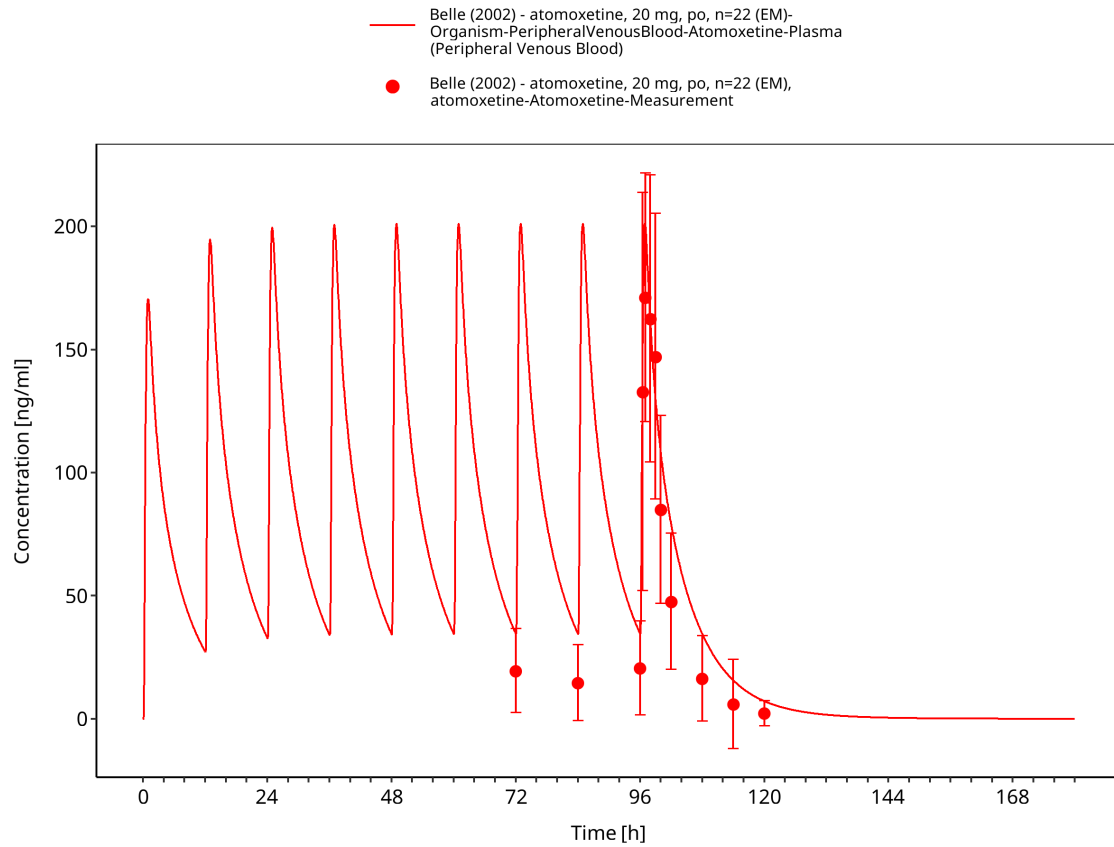


**Figure 3-7: Time Profile Analysis**

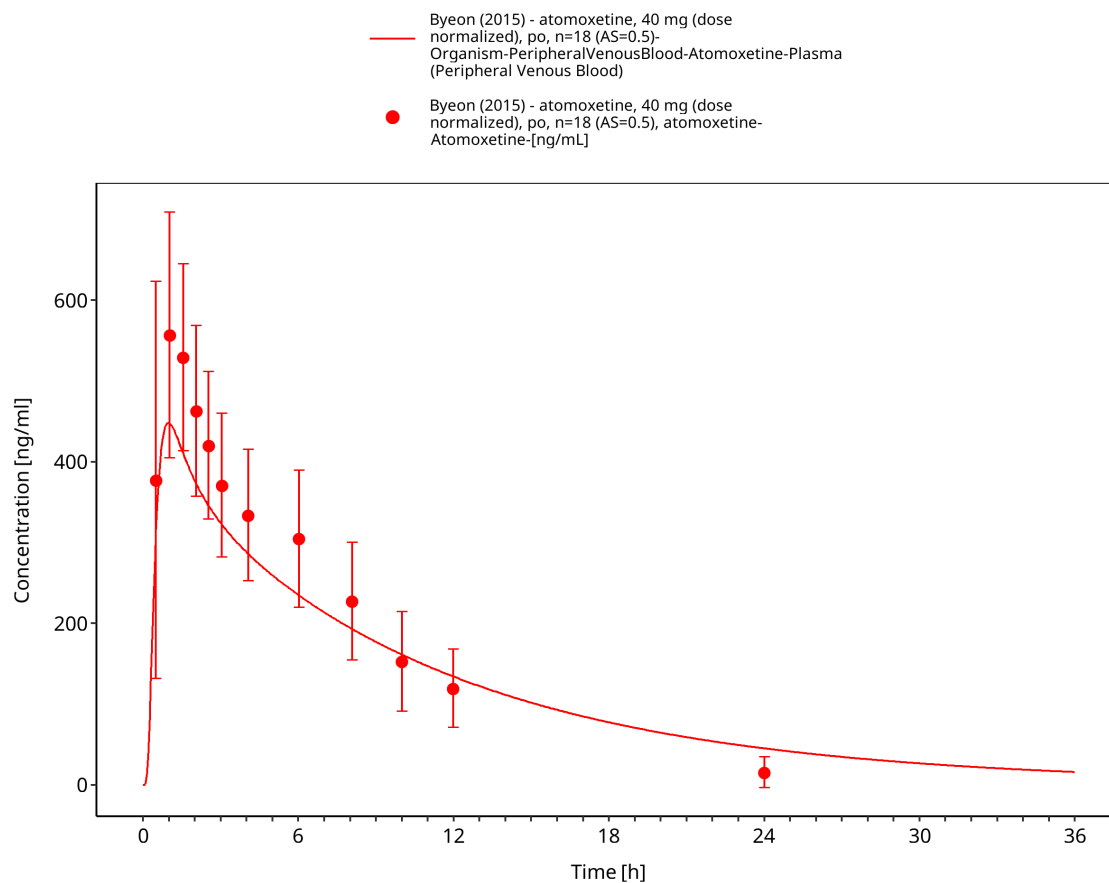


**Figure 3-8: Time Profile Analysis**

### 3.3.2 Model Verification

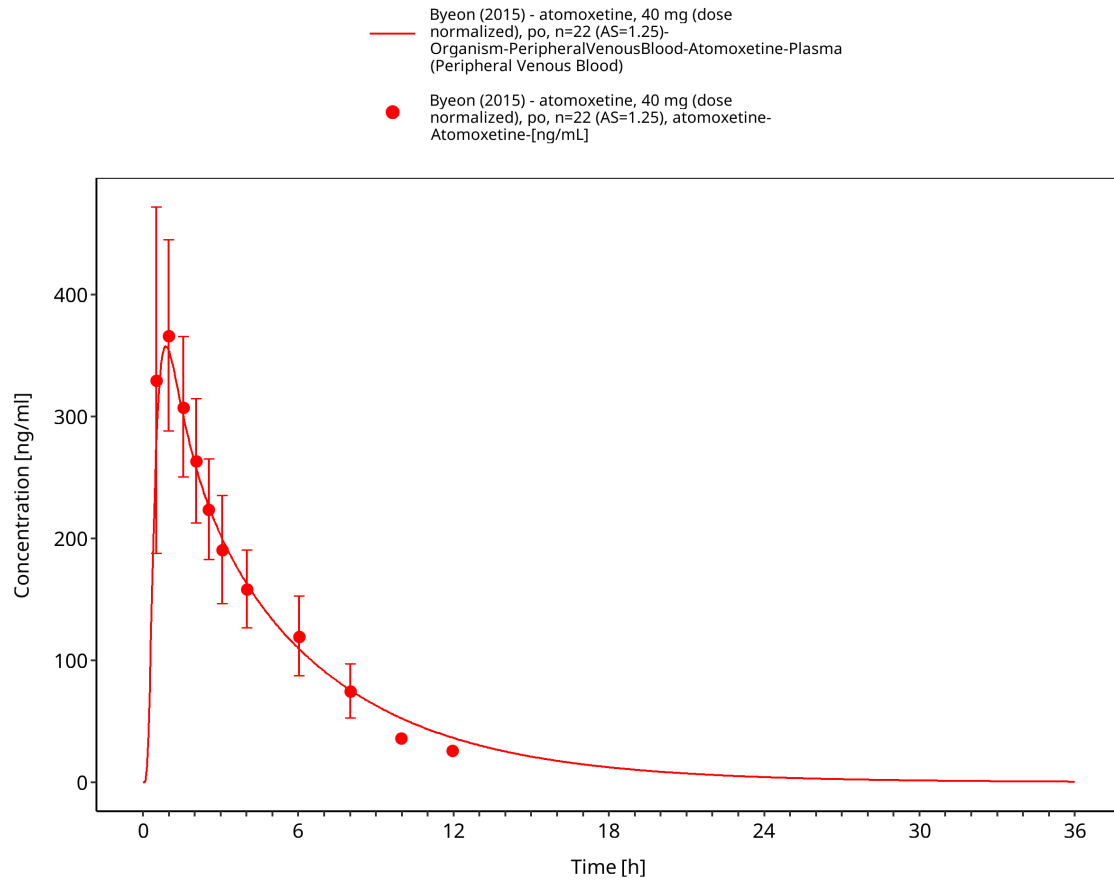


**Figure 3-9: Time Profile Analysis**

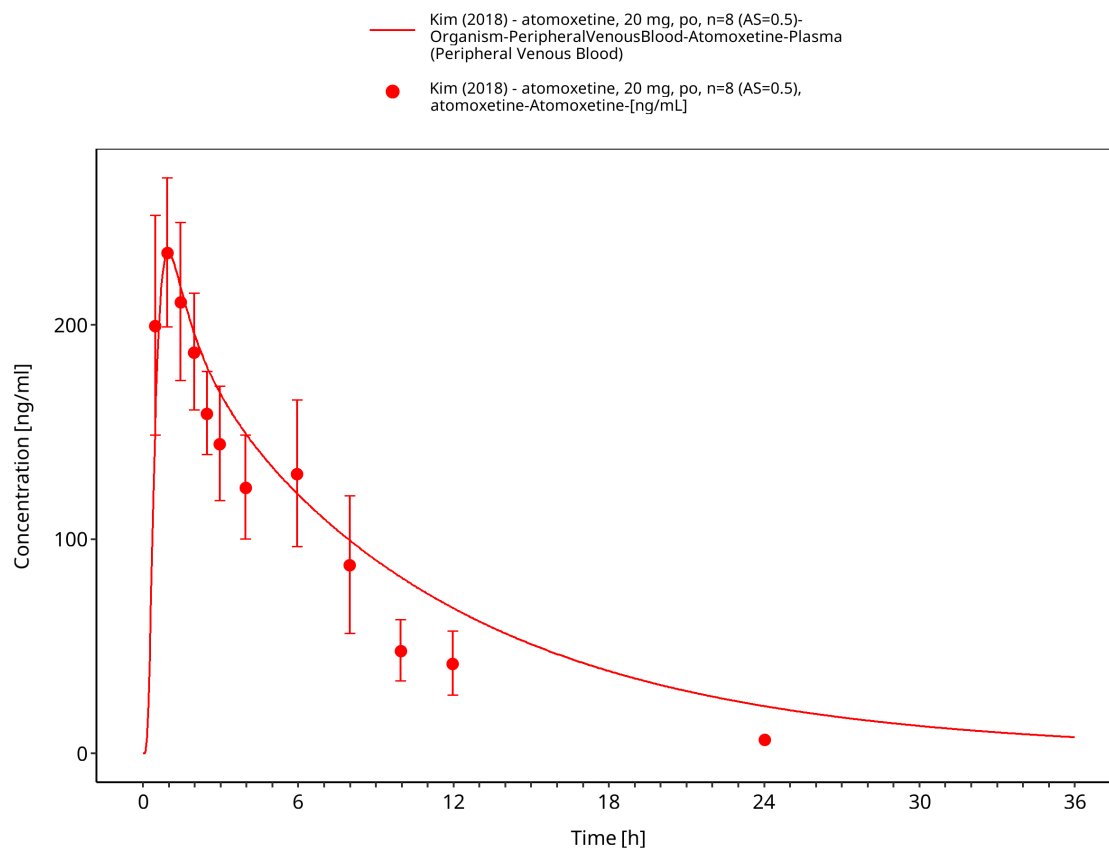


**Figure 3-10: Time Profile Analysis**

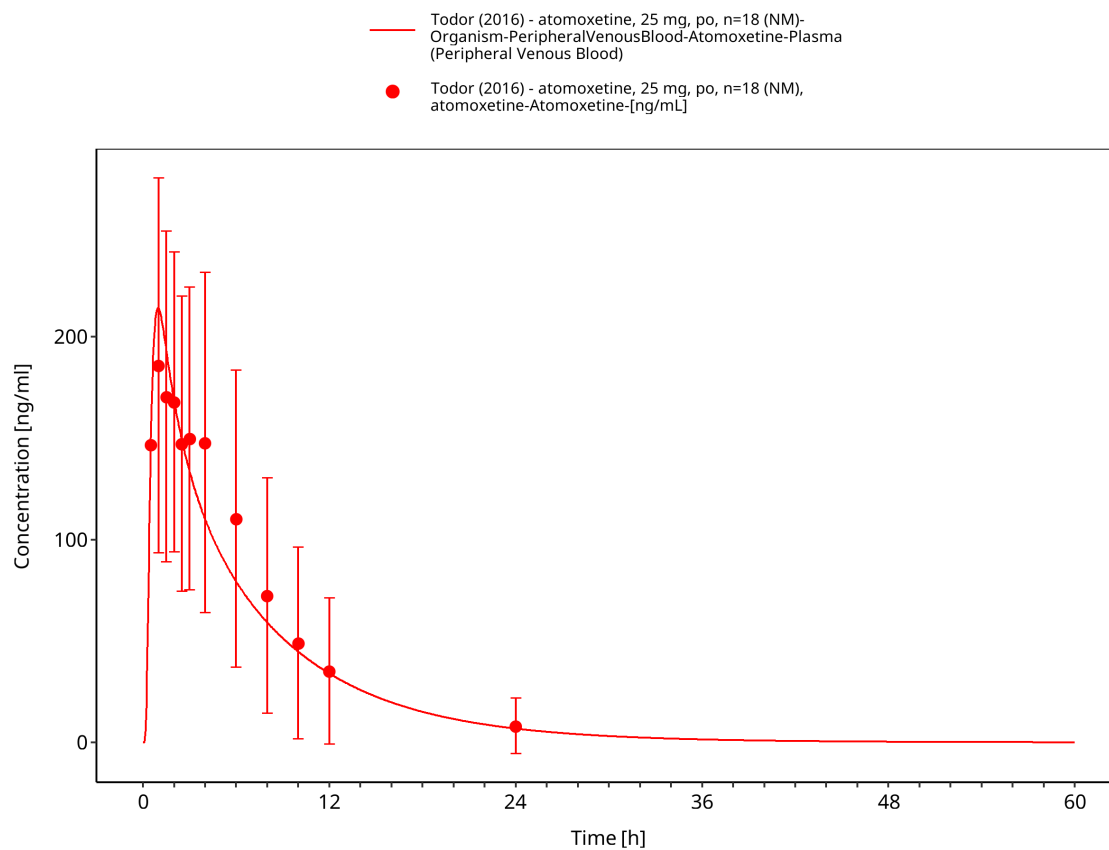




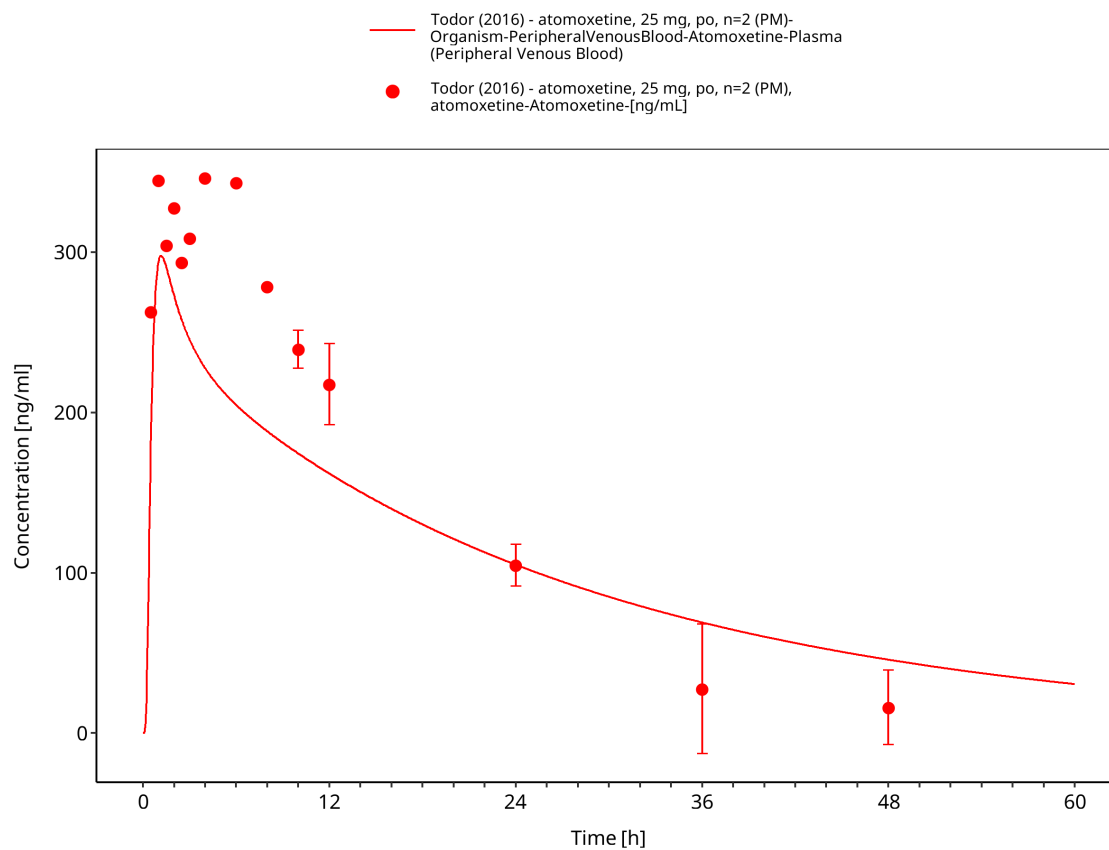
**Figure 3-11: Time Profile Analysis**



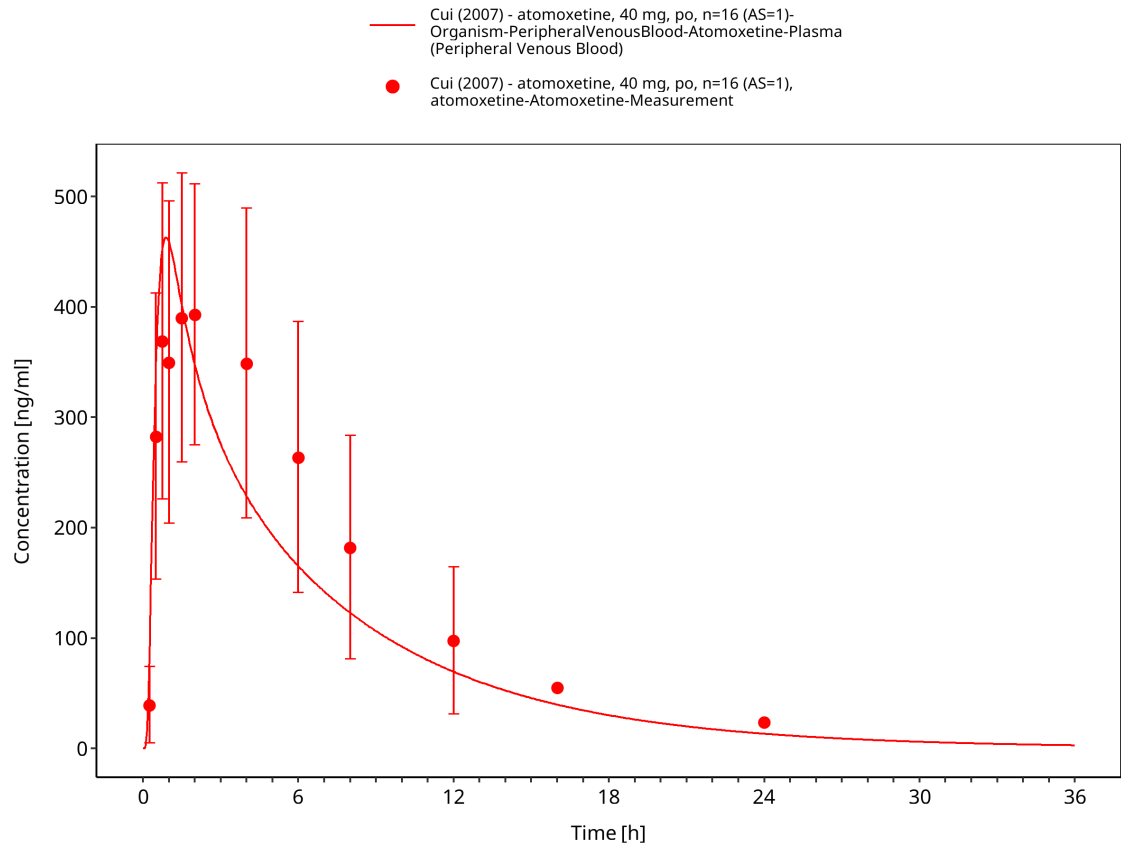
**Figure 3-12: Time Profile Analysis**



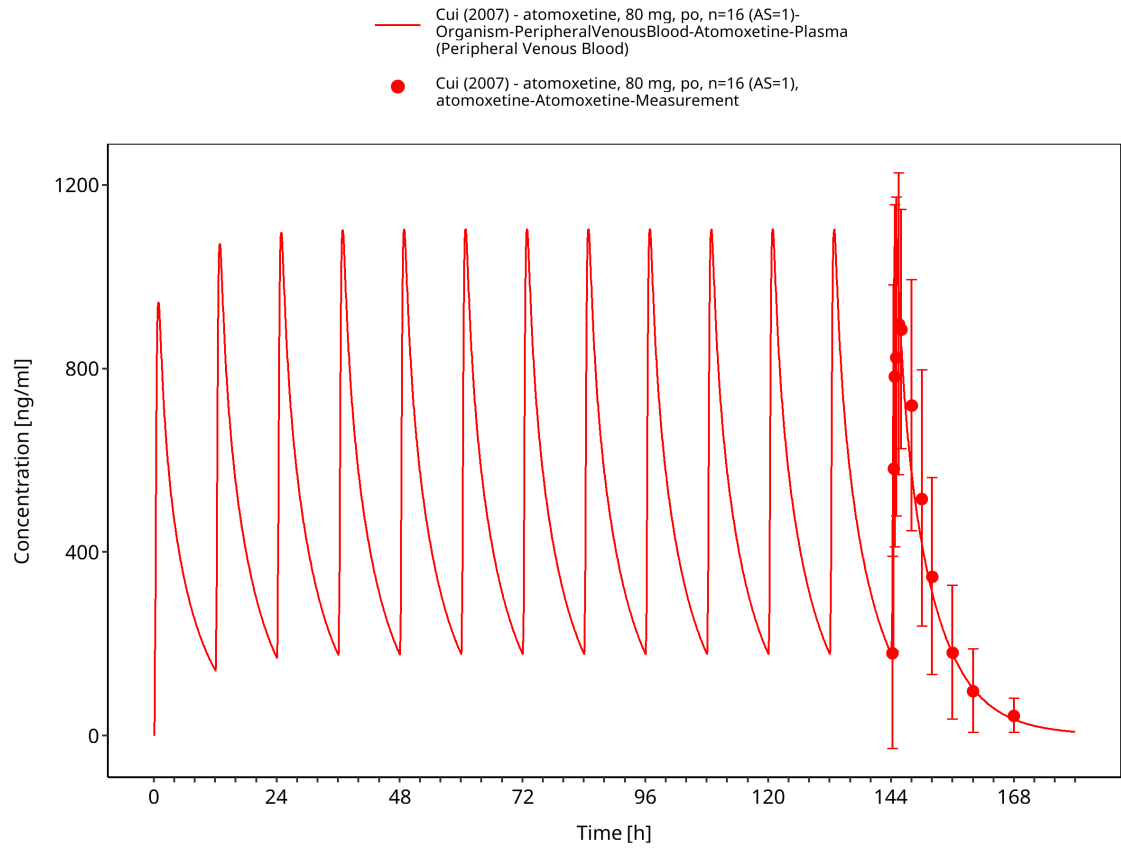
**Figure 3-13: Time Profile Analysis**



**Figure 3-14: Time Profile Analysis**



**Figure 3-15: Time Profile Analysis**



**Figure 3-16: Time Profile Analysis**

## 4 Conclusion

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The presented PBPK model adequately describes the oral pharmacokinetics of atomoxetine in the evaluated adult CYP2D6 phenotype and activity-score groups.

## 5 References

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